

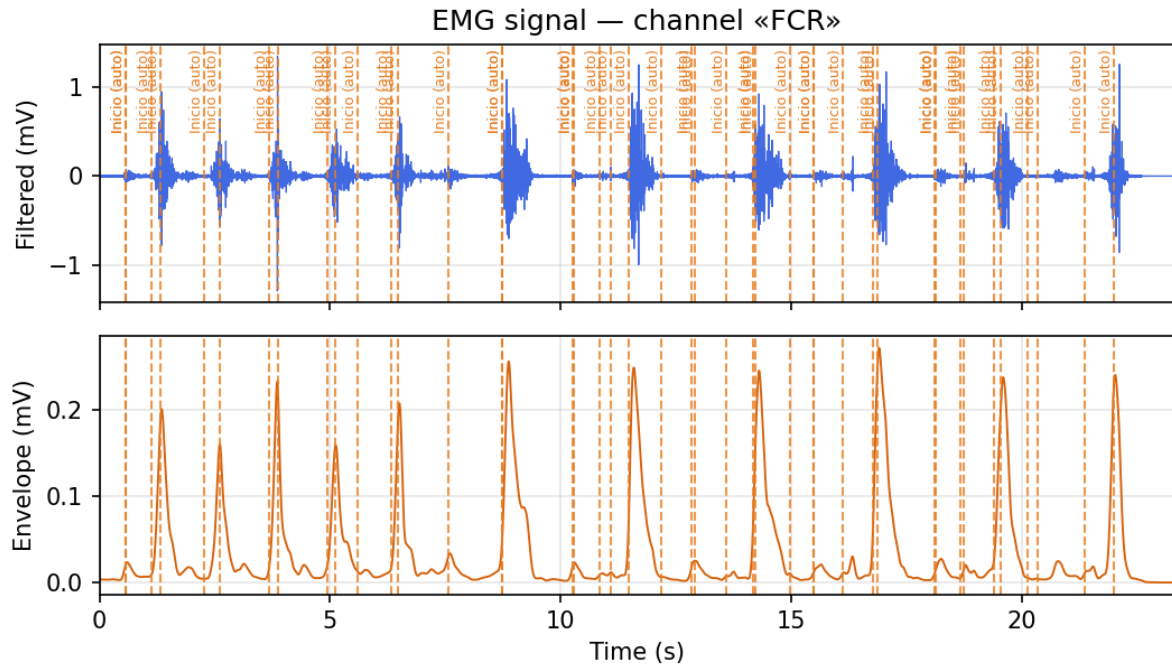
EMG recording and analysis report

Generated on: 2026-09-26 12:26

Test identifier: P01

File: ejemplo.edf

Protocol: Agonist / antagonist contraction



Metrics

Metric	FCR	ECR
Duration	23.3 s	
Analysed window	116.70-140.00 s of 140.0 s	
Global RMS	0.0876 mV	0.0526 mV
Mean frequency (MNF)	142.3 Hz	106.9 Hz
Median frequency (MDF)	126.5 Hz	93.1 Hz
iEMG	0.806 mV·s	0.617 mV·s
Fatigue evidence	Not conclusive — the trend does not fit (+0.22 Hz/s, $R^2=0.01$). Fatigue needs a contraction held long enough for the trend to show.	Not conclusive — the trend does not fit (+0.18 Hz/s, $R^2=0.00$). Fatigue needs a contraction held long enough for the trend to show.

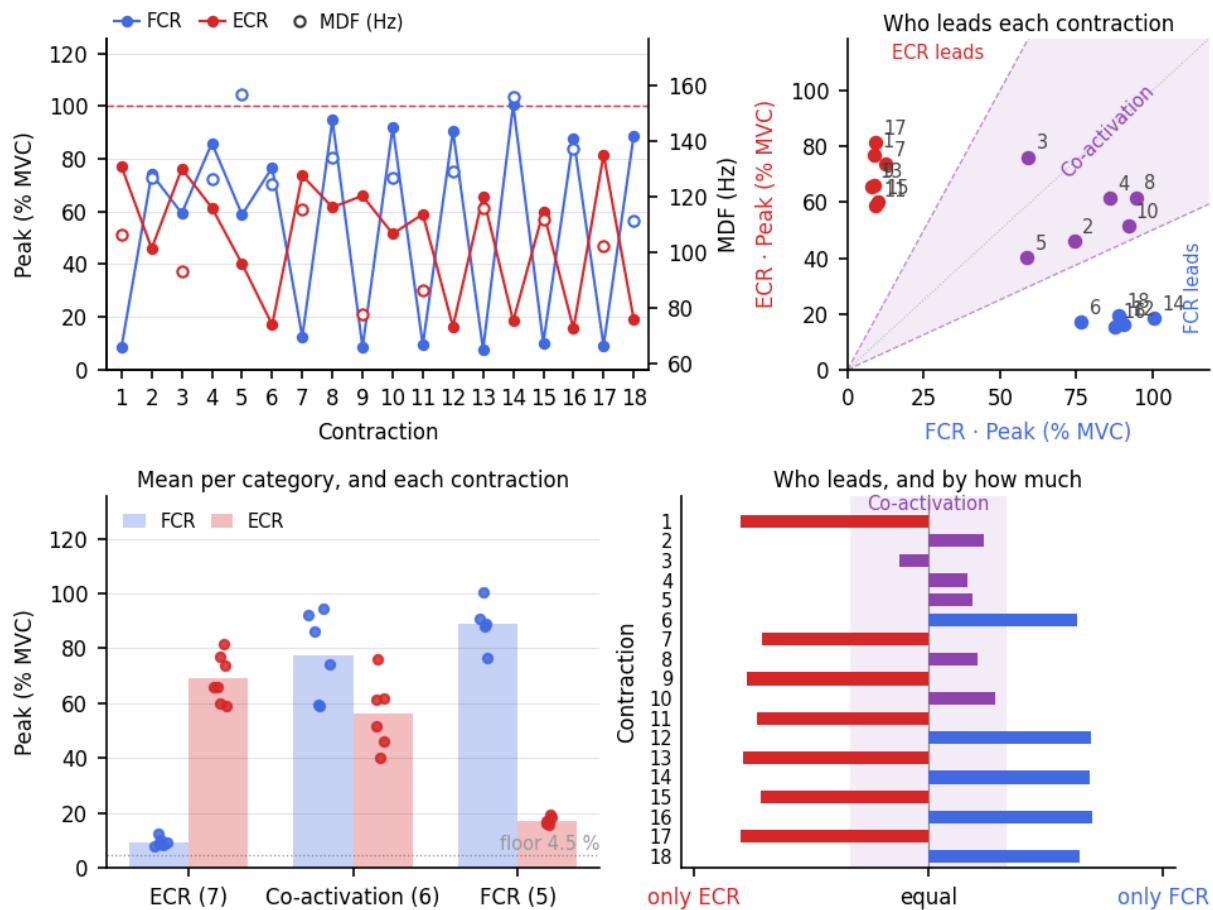
Calibration (maximal voluntary contraction)

Muscle	Reference	Source	Task maximum
FCR	0.271 mV	calibration in this recording (6 repetitions)	101 % MVC (envelope peak)
ECR	0.197 mV	calibration in this recording (6 repetitions)	82 % MVC (envelope peak)

Calibration repetitions

Muscle	Repetition	Value	Other muscle during it
FCR	1	0.271 mV	29 %
FCR	2	0.220 mV	21 %
FCR	3	0.193 mV	22 %
FCR	4	0.166 mV	27 %
FCR	5	0.157 mV	22 %
FCR	6	0.200 mV	18 %
ECR	1	0.160 mV	10 %
ECR	2	0.138 mV	10 %
ECR	3	0.197 mV	17 %
ECR	4	0.159 mV	17 %
ECR	5	0.178 mV	14 %
ECR	6	0.163 mV	9 %

Contractions



#	Start (s)	Duration (s)	Muscle	RMS (mV)	Peak (% MVC)	MDF (Hz)
1	0.5	0.42	ECR	0.119	77	107
2	1.2	1.02	Co-activation	0.111	74	127
3	2.4	0.95	Co-activation	0.076	76	93
4	3.7	0.97	Co-activation	0.119	86	126
5	4.9	1.05	Co-activation	0.082	59	157
6	6.3	0.54	FCR	0.133	77	124

#	Start (s)	Duration (s)	Muscle	RMS (mV)	Peak (% MVC)	MDF (Hz)
7	7.2	0.63	ECR	0.097	74	115
8	8.6	0.75	Co-activation	0.172	95	134
9	10.2	0.52	ECR	0.088	66	78
10	11.4	0.65	Co-activation	0.189	92	127
11	12.8	0.55	ECR	0.082	59	86
12	14.1	0.75	FCR	0.163	91	129
13	15.4	0.57	ECR	0.105	66	116
14	16.7	0.75	FCR	0.190	101	156
15	18.1	0.54	ECR	0.084	60	112
16	19.3	0.72	FCR	0.158	88	137
17	20.1	1.20	ECR	0.083	82	102
18	21.5	1.12	FCR	0.121	89	111

Co-activation (Falconer-Winter)

Whole recording — accept the fragments for one value per window

Window	FCR — Mean activation (% MVC)	ECR — Mean activation (% MVC)	Co-activation index
Whole recording (0.0–23.3 s)	12	10	31 %

Configuration used

Parameter	Value
Sampling rate	1000 Hz
Channel	FCR, ECR
Band-pass	20.0-450.0 Hz
Notch (mains)	50.0 Hz
Envelope (low-pass)	5.0 Hz
RMS window	50.0 ms
Detection sensitivity (k)	4.4
Device	BITalino